

Deep-learning based autonomous-exploration for UAV navigation

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ABSTRACT

Unmanned Aerial Vehicles (UAVs), functioning as aerial robots, undertake multifarious missions within intricate and uncharted environments, necessitating autonomous exploration for efficient UAV navigation. This paper introduces a Deep Learning-based Autonomous UAV Exploration method (DLAE), which distinguishes itself from traditional autonomous ground robot exploration by employing cameras instead of radar sensors. This difference propels us toward proposing a hybrid action space that integrates both positional and yaw actions, aimed at circumventing the limitations imposed by the Field of View (FOV). Furthermore, to enhance learning efficiency in autonomous UAV exploration, we design an invalid action masking scheme. Significantly, our contribution includes the development of an autoregressive network model alongside a distinct training procedure specifically tailored for autonomous UAV exploration. Comprehensive experiments conducted across a variety of test maps reveal that our proposed method surpasses existing learning-based approaches, delivering superior exploration efficiency and reduced decision-making time in comparison to conventional strategies. Moreover, physical simulations have corroborated the effectiveness of our method in practical scenarios, showcasing its potential for real-world applications.

1. Introduction

Autonomous exploration is essential for robotic autonomous navigation without prior environmental knowledge, as in unknown environments, robots cannot execute motion planning and must explore the environment first. In unknown environments, autonomous exploration helps robots to explore an area more efficiently and quickly. However, each type of robot has a relevant exploration method. Unmanned aerial vehicles (UAVs), as a type of flying robot, have excellent flexibility and can enter complex rooms for autonomous exploration. Moreover, the ground environment does not hinder the UAVs' autonomous exploration process, as by overcoming obstacles they can establish more comprehensive and detailed three-dimensional maps. Overall, employing UAVs for autonomous exploration has a good prospect and the technology has a wide range of applications [1], including rescue and reconstruction of unknown environments.

However, using UAVs for autonomous exploration imposes greater demands on methods, such as requiring a smooth and continuous trajectory during flight. It should be noted that most autonomous exploration methods are designed for ground robots, which are more appropriate for sudden stops and have relaxed requirements for trajectory continuity. Opposing, discontinuous and unsmooth trajectories lead to UAV tracking difficulties and cause damage. Moreover, due to endurance limits, autonomous exploration methods for UAVs must consider the power consumption of exploration paths. Unlike ground robots, UAVs

consume energy while hovering and making a decision. Hence, an autonomous exploration method with faster decision-making can save more energy. Furthermore, limitations considering the UAVs' load and structure add an extra challenge for the UAVs to carry omnidirectional exploration equipment. Therefore the pose and field of view (FOV) should be considered during the exploration. Therefore, overall, the problem of autonomous UAV exploration imposes higher requirements for the time and efficiency performance of the autonomous exploration method employed.

Spurred by the challenges presented above, this paper proposes a deep-learning based autonomous exploration method (DLAE). Generally, autonomous exploration methods can be divided into two categories: traditional methods based on frontiers [2] and deep-learning-based methods [3]. Typically, the former predicts multiple frontier points and selects the one that maximizes the exploration. The weakness of this method is that a large number of frontier points must be walked through, significantly increasing the computational burden. The latter, by replacing the selection with the network forward inference, a continuous process defined as a Markov decision process [4], significantly improves the decision-making efficiency. Overall, deep reinforcement learning (DRL) algorithms can find better policies for global reward.

Large action spaces and invalid actions are the major problems in autonomous exploration, as it is challenging for deep reinforcement

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learning algorithms to explore an effective policy. A common approach is to discretize the action space [3]. However, FOV limitations increase the number of action spaces, as UAVs must change the camera orientation during explorations, increasing the difficulty of searching for the solution. Besides, agents should refrain from selecting locations within obstacles or unexplored areas during exploration, as such actions are deemed invalid and could lead to unsafe maneuvers or a lack of progress. A typical approach is to provide a negative reward for these invalid actions [3,5]. Nevertheless, increasing the negative rewards prohibits the robot from learning a great policy [6]. Hence, designing the action space and avoiding invalid actions positively impacts the agents. This is the key to solving the autonomous exploration problem.

This paper focuses on the problem of autonomous UAV exploration policies in unknown environments. Our main contributions are summarized as follows.

- (1) We propose a deep-learning-based autonomous exploration method (DLAE) for UAV navigation in unknown environments. Our method achieves faster decisions and more efficient exploration planning than traditional methods.
- (2) We design a hybrid action space to describe autonomous UAV exploration. By separating the position and pose action, we deal with the FOV limitation in UAV exploration.
- (3) We develop a masking scheme to filter out invalid actions in autonomous exploration, enhancing UAV safety and the efficiency of deep learning methods. This approach proves more effective than using negative rewards.
- (4) We develop an autoregressive discretization network and a separation training process to solve the problem of excessive action space. The special network and training process reduces the difficulty of policy learning with excessive action space, affording a better result than other DL-based methods.

2. Related work

The literature suggests several methods for autonomous exploration [7–9], with the autonomous exploration policies divided into traditional and intelligent methods. The former has a long research history [2] and a wide range of practical applications. The latter methods [3,7,9] use neural networks as a policy, which can be improved through learning.

Traditional methods typically perform area exploration by leveraging frontiers [2], defined as regions demarcating the boundary between known free space and areas yet to be explored. By advancing towards these new frontiers, a robot can progressively expand its mapping into previously uncharted territories, thereby systematically exploring the entire environment. These approaches prioritize the identification and selection of optimal frontier points as vantage points for observation. Some works use information gain [10–13] to maximize the accuracy of the map-building process, while others [14] employ particle filters to conduct accurate exploration, mapping, and localization. For instance, in [8,15], the authors conducted autonomous exploration with unmanned aerial vehicles by rapidly-exploring random trees and dynamic exploration planners. In [16], researchers also utilize receding horizon and branch search to identify the next best view (NBV) point. Additionally, existing works have demonstrated that the efficiency of selecting candidate points is significantly enhanced by applying Bayesian optimization and Gaussian estimation. [17] improves the efficiency of selecting candidates. Although researchers have noted the field of view in exploration [18], traditional methods impose a significant computational burden to traverse the frontier candidates and thus increase the time to make decisions. However, traditional methods incorporate various concepts, such as occupancy maps, information entropy, and Bayesian optimization, inspiring the research toward intelligent methods.

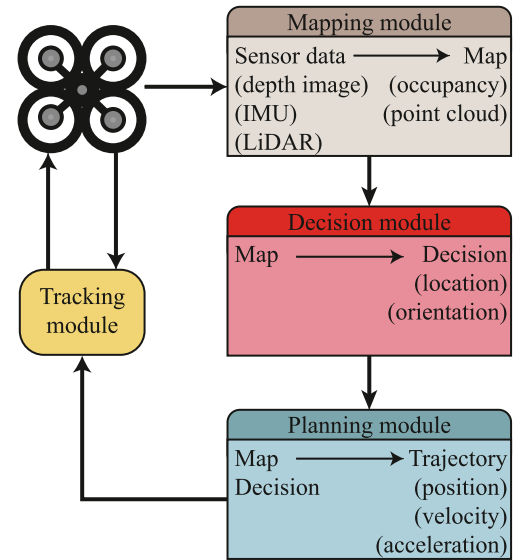


Fig. 1. Autonomous UAV exploration in unknown environments. The mapping module utilizes information from the UAV sensors and reconstructs the map. The decision module decides the sensing action from the map and the planning module creates a flight trajectory on the map to achieve the sensing action. The tracking module tracks the planning trajectory and sends the control commands to the UAV.

Intelligent methods typically employ neural networks as a policy, e.g., DRL algorithms assist neural network policies to make better decisions through learning from the environment. Currently, researchers consider autonomous exploration as a decision sequence problem, and given the development of deep reinforcement learning, neural network policies have been used to solve decision sequence problems, which have already reached the human level capabilities in many fields [19]. Existing intelligent methods can use RGB images as input to provide an end-to-end policy [20,21], while other works construct a supervised learning problem [22] to learn autonomous exploration policies. Alternative solutions, such as Actor-Critic(A3C) [23], have demonstrated that policies can achieve better results using a more advanced model. Overall, neural networks can replace many parts of the autonomous exploration process, including mapping [24], decisions [3], and controlling [9]. Although network policies are easier to transfer as a decisions module to the physical robot [3], most intelligent methods are based on radar and ground robots, there are no intelligent studies utilizing neural networks that investigate the limitations of FOV and the flight characteristics of UAV.

3. Problem statement

Autonomous UAV exploration in unknown environments is a complex task that requires multiple modules to collaborate (Fig. 1). A complete exploration strategy comprises a module mapping sensor data to occupancy maps, a policy model for global decisions based on map data and UAV pose, a planning module to plan the flight trajectory [25, 26], and a tracking module to track the trajectory. Multiple modules are closely linked to complete the Autonomous UAV exploration mission and therefore must be designed with holistic autonomous exploration strategies.

3.1. Mapping module

Autonomous exploration aims to build a complete map of unknown environments, with mapping involving process and goal attributes for the mission. Specifically, the decision module selects the action that maximizes the mapping efficiency and the planning module requires

map data to avoid obstacles. For the latter task, the UAVs acquire information about their environment through their onboard sensors and use mapping modules to convert this information into occupied maps required for the other modules. The function of the mapping modules is defined as

$$m_t, l_t = f_{\text{mapping}}(l_{t-1}, m_{t-1}, data_t), \quad (1)$$

where m_t, l_t are the grid map and estimated poses by the mapping module at t , respectively, and $data_t$ represents the sensor information acquired at t , which includes radar, RGB image, depth image, and IMU.

FOV limitations exist in UAV mapping. Indeed, unlike ground-based robots that carry an omnidirectional active radar, UAVs typically carry a monocular or depth camera with a certain angle of view because of the UAVs' load and structure constraints. Besides positions, policies must plan the UAV's pose to improve mapping efficiency.

3.2. Decision module

The decision module determines the next sensing action for the UAVs, which is determined based on the occupancy map and UAV pose. Since a UAV flies into a new position to continue exploration, the decision module must ensure that the position of the sensing action is within a safe region. Moreover, sensing action can maximize exploration. Hence, unlike ground robots that use radar, UAVs must also plan their poses to overcome FOV limitations. The decision module is modeled as

$$g_t, \theta_t = f_{\text{decision}}(l_t, m_t), \quad (2)$$

where g_t, θ_t are the action position and action pose given by the decision module. The decision module uses the current UAV position l_t and the map data m_t to select the action.

Autonomous exploration in unknown environments is a continuous process, during which UAVs have to make periodic decisions until all exploration goals are achieved. Therefore, selecting the appropriate actions needs close attention to afford objective continuity and maximize global rewards.

This paper focuses on developing decision modules for autonomous exploration by addressing some problems that exist in the current UAV decision modules. Specifically, we propose a DRL-based approach that affords a better sequential performance and parallel computing capability than traditional methods [2,17]. Compared with other DRL methods, our method can effectively handle the problem of limited FOV and invalid actions.

3.3. Planning module and tracking module

The planning module determines flight trajectories based on decisions and available map data to facilitate sensing actions. Therefore, it is imperative that the module accounts for any obstacles within the planned trajectories to ensure safety. Moreover, the modules must plan smooth trajectories that can be tracked well by UAVs in practice. The planning module can be described as

$$\tau_{T_t:T_{t+1}} = f_{\text{planning}}(g_t, \theta_t, m_t, l_t), \quad (3)$$

where τ is the planning trajectory. It should be noted that unlike the decision step t , T_t represents the actual time at the decision step t . Since the planned trajectory should escape obstacles and unknown regions, the trajectory quality does not only rely on the trajectory planning module but also the action of the decision modules.

Many graph search methods can be used to find paths, such as the A* algorithm [27]. However, such paths are difficult to use by UAVs in practice, and therefore smooth trajectories are needed, guaranteeing continuous positions, velocities, and accelerations. For the planning method developed we employ an online safe trajectory generation method [28] that relies on fast marching and Bernstein-based polynomial. The developed planning module generates safe and efficient

trajectories for UAVs. It is noteworthy that there are numerous noteworthy studies concerning expedited local path planning in unknown environments [29,30]. However, active exploration tasks, such as scene reconstruction, are more precise procedures. Therefore, the rapidity of planned trajectory has less impact on active exploration.

The tracking module's goal is tracking planning trajectories physically, with the corresponding module described as

$$u = f_{\text{following}}(\tau_{T_t:T_{t+1}}, l). \quad (4)$$

The tracking module calculates a controlled quantity u based on the planned trajectories and the UAV's position and pose. The literature suggests many control theory methods for controlling the flight, e.g., classical PID control. In this paper, we use a more accurate linear quadratic regulator as the tracking module [31], which guarantees a smooth and accurate tracking process.

4. Deep reinforcement learning-based decision

This section presents a DRL-based policy for autonomous UAV exploration. Our main work focuses on enhancing the DRL methods and adapting them to the autonomous UAV exploration problem.

4.1. Objective function

The core question in an exploration policy is how to efficiently obtain information about the environment. Considering the UAVs' battery capacity, we propose an objective function for the exploration, which aims to complete the exploration with minimal battery consumption. The objective function can be written as

$$O(m_{t_0:t}) = \Delta H(m_t) - \int_{t_0}^t P \times dt. \quad (5)$$

Inspired by the Shannon entropy, we evaluate the exploration [10] degree based on the Shannon entropy increment. The Shannon's entropy over an occupancy grid map m is defined as:

$$H(m) = \sum_{i,j} p(m_{i,j}) \log p(m_{i,j}). \quad (6)$$

Although it is hard to count the amount of energy consumed during UAV exploration, because of the tracking module, the length of the planned trajectories almost determines the UAVs' energy consumption. Hence, we regard the trajectory length as a substitution of energy consumption, where the length can be easily calculated. The flight length and energy consumption are defined as follows:

$$L_{t_0:t} = \int_{t_0}^t (|v_x| + |v_y| + |v_z|) \times dt, \quad (7)$$

$$\int_{t_0}^t P \times dt = k \times L_{t_0:t}, \quad (8)$$

where v_x, v_y, v_z denote the velocity decomposition on the x, y and z -axis. In this paper, the trajectory length is calculated by integrating the UAV's velocity over time. This process aims to reduce the Shannon entropy of the occupied grid map and thus plan a short exploration path.

Planning modules generate UAV trajectories to achieve sensing actions from decision modules. Our planning module involves an online safe trajectory generator [28], which incorporates three steps. First, the planning modules obtain a feasible path to the sensing action position through graph search methods. To optimally leverage the identified free space for planning the most efficient flight route, our algorithm extends the shortest feasible, albeit non-smooth, trajectory on the occupancy map to generate a sequence of Box blocks. These blocks are designed to encapsulate the UAV's initial and destination points within the first and last blocks, respectively, while guaranteeing intersections between adjacent blocks to ensure continuity. Additionally, each block delineates a zone deemed safe for navigation. Meanwhile, each block will

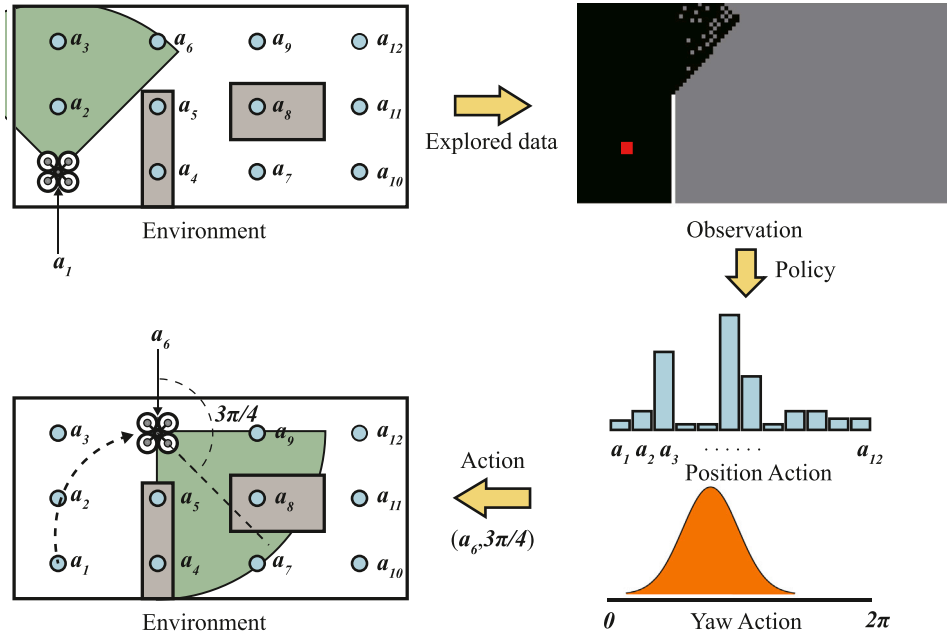


Fig. 2. Environment and Action Space. In the depicted environment, blue points indicate permissible actions, distributed uniformly in accordance with map dimensions and the overall action count, while grey areas signify obstacles. The observation integrates the UAV's location and the occupancy grid map, represented as an RGB image. The policy deduces action distributions from these observations, samples from these distributions, and communicates the selected action to the environment.

enlarge in space to maximize the utilization of free space. The planning problem can be solved as a quadratic programming (QP) with corridor constraints. However, instead of the traditional monomial polynomial basis, we rely on the Bernstein polynomial basis. Furthermore, we employ Bezier curves to represent planning trajectories due to their special properties. It is worth noting that we search for trajectories with a minimal snap [32] instead of a minimal jerk because of the differential flatness properties of UAVs. This is important as a minimum snap trajectory will save energy during tracking, while generating flight trajectories from constraints is similar to [28]. Moreover, the process from decision to flight trajectories will be emphasized in this paper.

Algorithm 1 UAV decision-making and trajectory generation

Input: θ : Decision Policy; m : Occupancy grid map; p_t : Pose of UAV at moment t

Output: $trac$: UAV trajectories of position, velocity, and acceleration

- 1: Policy θ decide next sensing action pose p'_{t+1} based on m and p_t
- 2: **if** No viable path from p_t to p'_{t+1} **then**
- 3: Return to current position
- 4: **else**
- 5: Finding a safe path $trac_0$ from p_t to p'_{t+1}
- 6: **end if**
- 7: Initialize a flight corridor set $\mathbf{Box} = \{box_i, i \in N\}$ and enlarge it based on $trac_0$ and m
- 8: **if** The solution of QP with $\mathbf{Box}$ does not exist **then**
- 9: Return to current position
- 10: **else**
- 11: Calculate the set of control points $\mathbf{C} = \{c_j^i, i \in N, j \in M\}$ to minimum the snap of trajectory with $\mathbf{Box}$.
- 12: **end if**
- 13: Calculate UAV trajectories $trac$ from the set of control points $\mathbf{C}$.

Based on the objective function of the autonomous exploration, we define a reward function for the policy decisions as:

$$r(s_t, a_t) = \Delta H(m_t) - k \times \int_{T_t}^{T_{t+1}} P \times dt, \quad (9)$$

where a_t denotes the policy decision at state s_t . The reward function contains two parts, Shannon's entropy increment, and a flight's energy

consumption. Such a reward function helps the policy to decide on the most efficient action for the current state. In order to avoid the rewards being too small after completing most regions' exploration the agents will receive larger rewards after completing all region exploration.

In this work the agents are trained using policy gradient methods [33]. Compared with the Q-network methods, policy gradient methods have better learning and generalization capabilities. The goal of the policy gradient methods can be described as

$$U(\theta) = \sum_{\tau} P(\tau; \theta) R(\tau), \quad (10)$$

where θ denotes a policy network parameter, and $P(\tau; \theta)$, $R(\tau)$ are the probability and ultimate reward of trajectory $\tau(s_0, a_0, r_0, \dots, s_{T-1}, a_{T-1}, r_{T-1})$. The policy gradient methods optimize $U(\theta)$ through learning. It is easier to solve policy learning as an optimization problem and many interesting studies focus on policy gradient methods [34,35]. The proximal policy optimization (PPO) [35] is one of the most compact and efficient algorithms, aiming to avoid excessive updates during optimization. The optimization can be described as:

$$\theta_{k+1} = \arg \max_{\theta} \mathbb{E}_{s,a \sim \pi_{\theta}} (L(s, a, \theta_k, \theta)), \quad (11)$$

where L is a substitute for $U(\theta)$. In PPO, L can be described as:

$$L(s, a, \theta_k, \theta) = \min \left(\frac{\pi_{\theta}(a|s)}{\pi_{\theta_k}(a|s)} A^{\pi_{\theta_k}}(s, a), g(\epsilon, A^{\pi_{\theta_k}}) \right), \quad (12)$$

where

$$g(\epsilon, A) = \begin{cases} (1 + \epsilon)A & A \geq 0 \\ (1 - \epsilon)A & A < 0, \end{cases} \quad (13)$$

where $A^{\pi_{\theta}}(s, a)$ is the advantage function for the current policy that describes the improvement of a new policy over the old one. The advantage function is written as:

$$A^{\pi}(s, a) = Q_{\pi}(s, a) - V_{\pi}(s). \quad (14)$$

Furthermore, we improve policy learning and design a particular action space to fit autonomous exploration, assisting the UAVs to gain more rewards during explorations.

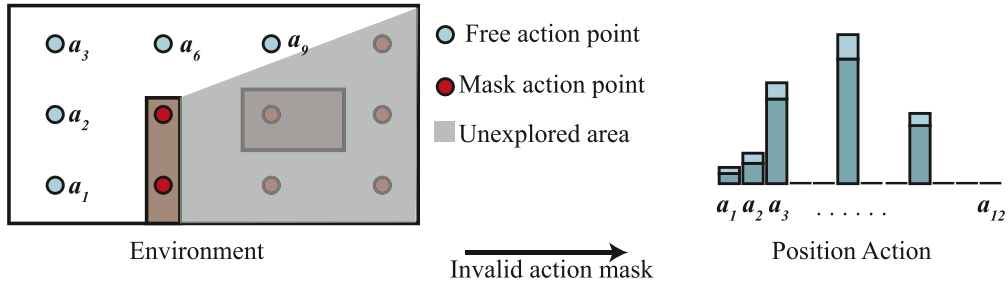


Fig. 3. Invalid action masking. Invalid actions are position actions in obstacles(brown area) and unexplored regions(grey area) marked in red and free position actions are marked in blue. The probability of choosing the masked invalid action is virtually zero. The policy will renormalize the probability distribution and obtain a new one.

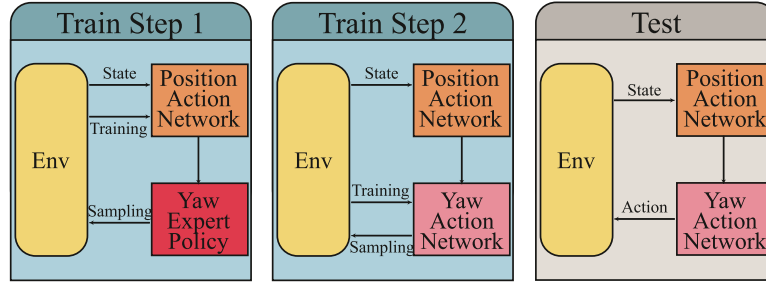


Fig. 4. Training step. In train step 1, the model obtains the state from the environment, with the position action network and yaw expert policy deciding sampling actions. During training step 2, the parameters of the yaw action network updated with a frozen position networks.

4.2. Action space and invalid action masking

One of the most significant challenges in DRL-based exploration is the FOV limitations, imposing agents to require position and yaw actions to make their decisions. Although a simple discretization is a common solution, is too expensive for position and yaw actions, as the growth of the action space impacts policy learning and makes it more challenging to learn a helpful one. Instead of simple discretization, we use hybrid action space and autoregressive discretization. As illustrated in Fig. 2, the UAV encapsulates the occupied map data and its positional information into an RGB image, serving as the current environmental observation. This observation, when processed by the policy through a neural network model, yields an action distribution encompassing both positional and yaw action distributions. These distributions indicate the UAV's probable next spatial positions and the corresponding yaw angles for upcoming explorations, respectively. The policy's output distribution informs the action selection for the subsequent step, utilizing sampling to specify the position point and associated yaw angle. Upon designation of the next exploration target, the environment orchestrates the UAV's trajectory and yaw adjustment, facilitating its flight along the planned path for exploration. This strategy affords the agents to explore an area based on position and yaw actions, effectively solving the FOV limitation of exploration.

Another challenge in DRL-based exploration is invalid actions. Indeed, when exploring unknown environments with UAVs some dangerous actions might be conducted, with policies setting the action position on an obstacle or inside unknown regions. These actions may cause damage due to collisions and therefore must be invalidated. A common solution is to give a larger negative reward [3] to invalidate such actions during training, forcing the model to learn to avoid such invalid actions. However, during an autonomous UAV exploration, the invalid actions might be vast, especially at the beginning of the exploration. Hence, it is difficult for agents to explore a trajectory with too many negative rewards. Therefore, we use the invalid action masking method to avoid executing invalid actions. Invalid action masking (Fig. 3)

empowers the agents to learn more efficiently and is a differentiable function described as follows:

$$\pi'(\cdot|s_t) = \text{softmax}(\text{inv}_s(l(s))), \quad (15)$$

$$\text{inv}_s(l(s))_i = \begin{cases} l_i & \text{if } a_i \text{ is valid in } s \\ M & \text{otherwise,} \end{cases} \quad (16)$$

where π' is the policy after masking and $\text{inv}_s(l(s))$ is a differentiable masking function that makes the gradient corresponding to the logits of the invalid action zero. And M is a large negative number. Masking works is effective with large and complex discrete action spaces, such as autonomous UAVs exploration.

4.3. Network architecture and training steps

Since setting positional actions and yaw actions in the same action space results in an overly large action space, it becomes difficult for the model to learn an effective policy. Therefore, we separated positional actions and yaw actions and controlled them with two different networks. As depicted in Fig. 5, this paper uses two neural networks with an Actor-critic (AC) structure to form the proposed DLAE model. The two networks are connected by the autoregressive discretization method, which links two networks, where the yaw action network input includes the observation and position actions. However, during training, the two networks can negatively influence each other, thereby increasing the complexity of the training process. In order to mitigate it, we propose a unique two-step training algorithm. As described in Fig. 4, during the first step, the algorithm solely trains the positional action network while the yaw action network is replaced with an expert policy. To prevent suboptimal parameters due to separate training, the expert policy explores all possible yaw actions and selects the action that yields the highest reward. Although this expert policy increases decision time, the issue will be resolved during the second training phase. After the first step, the algorithm freezes the parameters of the action network and optimizes the parameters of the yaw action network, this step is similar to let the yaw action network imitate the

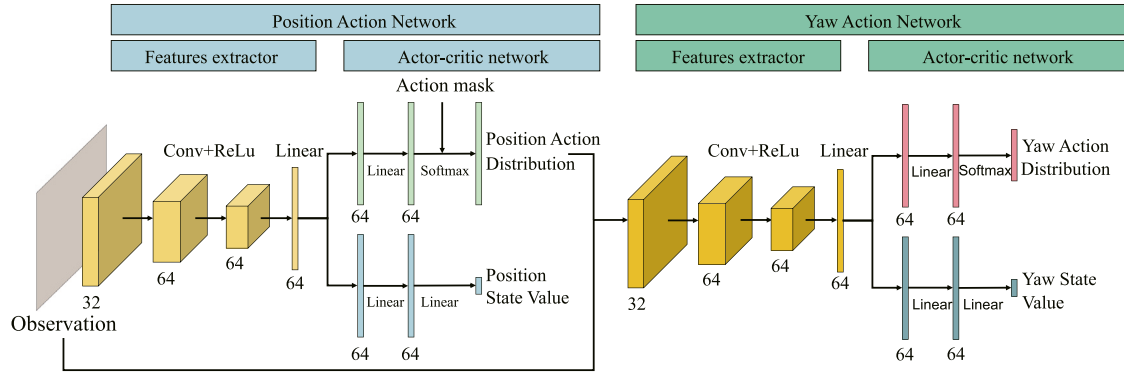


Fig. 5. Network Architecture of DLAE. The position network comprises a features extractor and an AC network. The yaw network is similar to the position action network but without masking.

previous expert policy, but it is able to reduce a significant amount of decision time. The experimental results and a comparison against other methods are presented in the next chapter.

5. Experiments results

This section challenges the proposed method's performance against other methods. All experiments use the same mapping, planning, and tracking modules except for the different decision modules. Considering the competitor methods, the proposed method is challenged against other DRL methods, and we analyze the training process and the test results. Then our method is compared with traditional frontier exploration methods, where we compare the exploration capability, efficiency, and computing time.

5.1. Evaluation metrics and experiments environments

The performance of all methods is evaluated based on the explored rate, explored length, decision time, and explored efficiency metrics.

The explored rate reveals the exploration completeness and is defined as the increment of the Shannon entropy divided by the initial value. An explored rate of 1 means that the map has been fully explored. The explored rate is defined as:

$$\text{Explored rate} = \frac{H(m_t) - H(m_0)}{H(m_0)}. \quad (17)$$

This paper uses the explored length to measure the UAV's energy consumption during the exploration. According to Eq. (7), the explored length is:

$$\text{Explored length} = L_{T_0:T_i}. \quad (18)$$

The decision time is the time that the decision module costs when the method finishes exploring the environment. The decision time is used to measure the time complexity, described as:

$$\text{Decision time} = \sum_{t=0}^N T_t. \quad (19)$$

The explored efficiency metrics are designed to highlight the capability of different methods to balance explored rates and explored time. An effective method explores more in less time and decides the right time to stop exploring. The explored efficiency is defined as:

$$\text{Explored efficiency} = \frac{\text{Explored rate}}{\text{Explored time}}, \quad (20)$$

$$\text{Explored time} = \frac{\text{Explored length}}{V_{limit}} + \text{Decision time}, \quad (21)$$

where V_{limit} is the maximum flight speed of the UAV. In this paper $V_{limit} = 1$ m/s.

Generally, a higher explored rate policy always costs more explored length. However, efficient policies balance this conflict with rapidly exploring unknown regions.

In order to verify our method's effectiveness and generalization, we trained it on a single map and tested it on four maps. Fig. 6 shows the details of training and testing maps.

5.2. Training analysis

This paper trains multiple DRL-based methods and the suggested DLAE on the training map and we analyze the training process and the results of the methods' exploration metrics on the test maps. The input to the network is an RGB image containing the UAV position information and occupancy grid maps, and the output of the network comprises a position and a yaw action. The position and yaw actions are implemented on the UAV through the planning and tracking modules.

Our method is challenged against some advanced DRL methods, i.e., DQN [36], PPO [35], and Mask PPO [6], with Fig. 7 illustrating the training reward curves. The total samples are 36 000, $\gamma = 0.8$, the learning rate is 0.0003, and the Clip range of DLAE, PPO, and Mask PPO is set to 0.3. Training takes around 12 h on a computer with RTX 3090 GPU and Intel Xeon E3-1275 3.8 GHz. Training DLAE involves two steps, as illustrated in Fig. 4, where the position action network is trained with a yaw expert policy in the first step and then the yaw action network is trained with the frozen parameters of the action position network. It is worth noting that DLAE and Mask PPO exhibit higher initial rewards compared to PPO and DQN. This is due to the fact that DLAE and Mask PPO do not select invalid actions for exploration purposes. Such an approach enables DLAE and Mask PPO to learn a good policy faster and better. Furthermore, DLAE has a perfect angle controller during the initial training phase, which allows the policy to focus more on learning the positional actions in active exploration.

Fig. 7 highlights that DLAE achieves the highest score on the training map, with DQN under-performing due to the large action space. Regarding PPO and Mask PPO, these achieve better results than DQN but still struggle to get a positive reward. Interestingly, during the initial training stages masking can help the policy to achieve better rewards than PPO. However, PPO soon learns a similar approach to Mask, and the two methods achieve very similar results. However, by training the model step-wise, DLAE reduces the difficulty of searching the action space and helps the policy to perform well.

5.3. Performance analysis

In this subsection, we compare the DLAE with the Proximal Policy Optimization (PPO) method [35], the Bayesian Optimization method [17], and Receding Horizon Next-Best-View method [16]. As illustrated in Fig. 8, Tables 1 and 2, the comparison includes four dimensions:

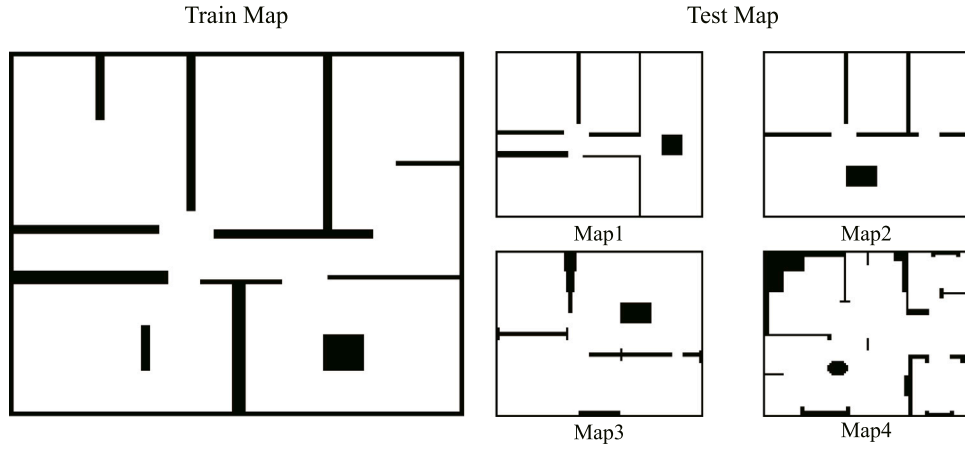


Fig. 6. Training and Testing Maps: Obstacles or walls are depicted in black. The Training Map, Test Maps 1 and 2 measure (8×10) meters each. Map 3 spans (6.4×8) meters, while Map 4 extends over (13.4×16.6) meters.

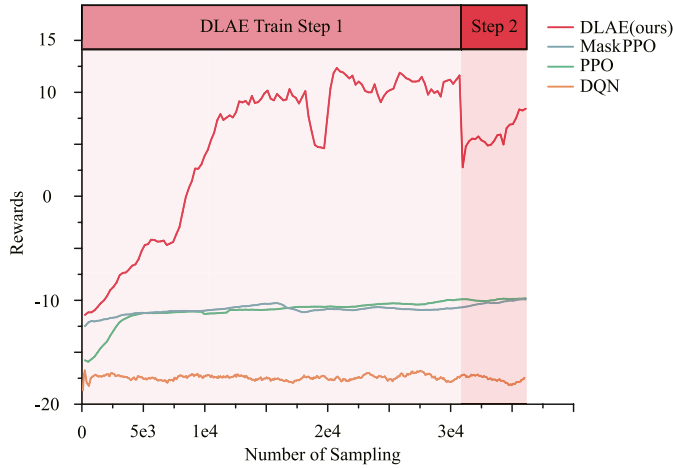


Fig. 7. Learning curves for exploration task. The total number of sampling for each method is 36 000.

Table 1
Decision time performance of different method on the test map.

Map	DLAE	PPO	NBV	BO
Map 1	0.02976	0.02597	3.13326	16.65234
Map 2	0.04642	0.03928	3.18955	8.59744
Map 3	0.05209	0.02961	5.57495	11.89902
Map 4	0.10529	0.02961	6.43902	19.51141

Table 2
Exploration efficiency of various methods on test maps.

	DLAE	PPO	NBV	BO
Map 1	0.035 ± 0.009	0.019 ± 0.002	0.02 ± 0.002	0.014 ± 0.001
Map 2	0.021 ± 0.002	0.016 ± 0.002	0.029 ± 0.007	0.021 ± 0
Map 3	0.038 ± 0.011	0.018 ± 0	0.024 ± 0.002	0.023 ± 0.001
Map 4	0.013 ± 0	0 ± 0	0.009 ± 0.001	0.009 ± 0.001
Mean	0.027	0.013	0.020	0.017

explored rate, explored length, explored efficiency, and decision time. Furthermore, the exploration paths of four methods on the map are showed in Fig. 9.

As Fig. 8, Tables 1 and 2 representing the advantages and characteristics of the four competitor methods. Concerning the exploration rate,

DLAE, NBV, and BO demonstrate exemplary performance, approaching 1 across all maps, albeit DLAE exhibits marginally lower rates compared to NBV and BO. Regarding exploration length, as illustrated in the figure 9, DLAE tends to select shorter paths to accomplish exploration tasks. Different from frontier based methods, the neural network-based method learns features such as rooms and obstacles in addition to sensing unknown regions. Learning the features of rooms and obstacles can help UAVs reduce explored length. UAVs relying on DLAE are more likely to scan a room at the door rather than fly deeper into it and are more likely to probe around obstacles rather than against them. These characteristics help UAVs to be more intelligent in autonomous exploration.

Subsequently, we compare the decision times for all methods. All experiments were performed on the CPU (without CUDA acceleration). One of our major findings is that replacing traditional methods with neural networks can significantly reduce decision times. Indeed, the computation time of deep neural network methods is significantly reduced compared to traditional methods. This improvement is significant for autonomous UAVs exploration, which means UAVs with DLAE can explore further while NBV and BO still make decisions, especially for indoor explorations. Due to the shorter paths and faster decision-making, DLAE provides a better result on explored efficiency than other methods, providing a more efficient and intelligent exploration solution that affords better continuity during DLAE decision-making.

In summary, DLAE achieves shorter explored lengths and improved explored efficiency by learning obstacles features. Moreover, with faster decision time, DLAE can perform well in autonomous UAVs exploration. For more experiments, the reader is referred to Appendix A.

5.4. Ablation study

In this study, we introduce a UAV autonomous exploration method leveraging deep learning, aimed at enhancing the model's adaptability to large action spaces and numerous invalid actions during UAV autonomous exploration training. We propose two specific approaches: the utilization of autoregressive models and invalid action masking within the exploration model. To examine the efficacy of these strategies in improving deep learning models for UAV autonomous exploration tasks, an ablation study was conducted (Table 3). This study focused on comparing exploration rates, path lengths, and efficiency across network models with varying configurations of these two methods.

Furthermore, to explore the influence of model parameter size on the performance of deep learning-based UAV autonomous exploration,

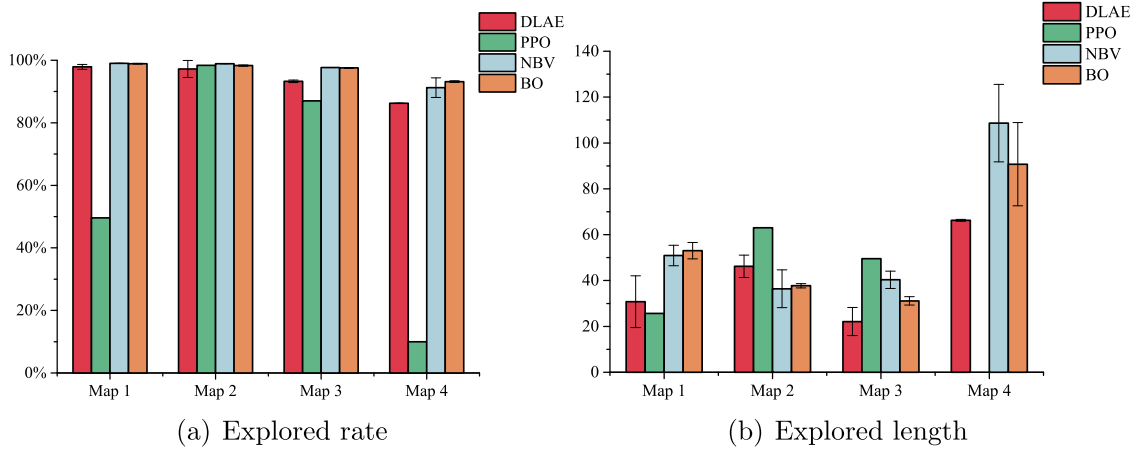


Fig. 8. Performance Metrics: Exploration Rate and Exploration Length on Test Maps. A higher Exploration Rate, approaching 1, indicates more thorough exploration. Conversely, a shorter Exploration Length denotes lower operational costs for UAVs to complete exploration tasks.

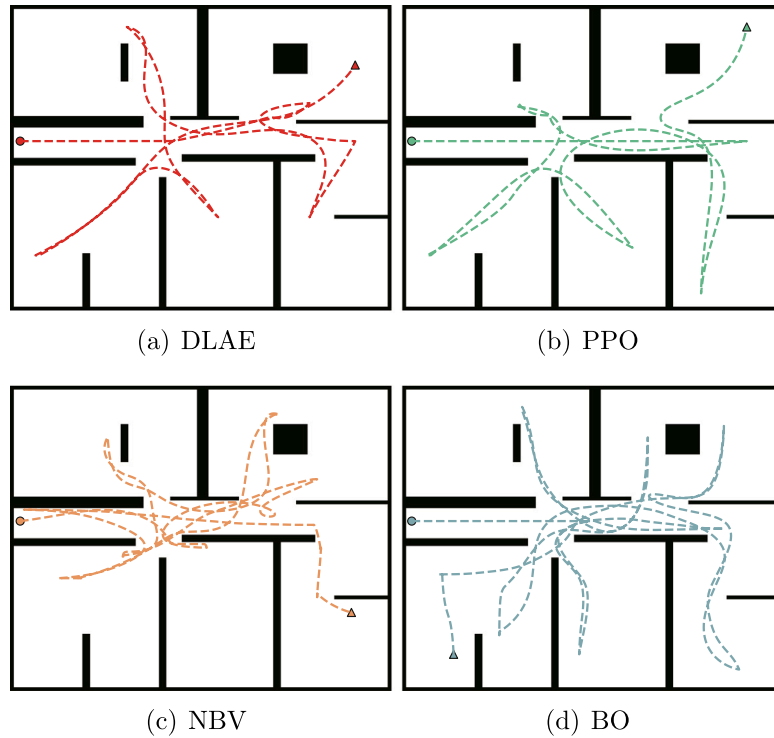


Fig. 9. Trajectory Visualization Simulation: The circle denotes the starting point, while the triangle indicates the endpoint of the trajectory. The explored length of DLAE is 45.63 m, PPO length is 48.82 m, NBV explored length is 55.1 m, and BO explored length is 72.59 m.

this paper presents the design and validation of a lightweight model. Although this model mirrors the structure and training methodology of the Deep Learning Autonomous Exploration (DLAE) model, it is distinguished by a reduced parameter count, offering insights into the balance between model complexity and exploration efficiency.

The outcomes of the ablation studies underscore that the methods introduced in this research markedly enhance performance over the baseline. Utilizing an autoregressive model not only elevates the exploration rate but substantively minimizes the required flight distance for comprehensive exploration, thereby streamlining path planning and boosting the UAV's operational efficiency. Moreover, the implementation of ineffective action masking has shown a substantial increase in the exploration rate within unfamiliar terrains, facilitating a more thorough exploration by the UAV. The integration of these two approaches

Table 3
Ablation study result.

Component	Choice				
Autoregression model	✓	✓	✓		
Invalid action mask	✓	✓		✓	
Model size	35.9 MB	2.08 MB	35.9 MB	21.7 MB	21.7 MB
Explore rate	0.94	0.79	0.637	0.87	0.58
Explored length	41.32	52.5	45.94	106.85	85.03
Explored efficiency	0.023	0.015	0.014	0.008	0.007

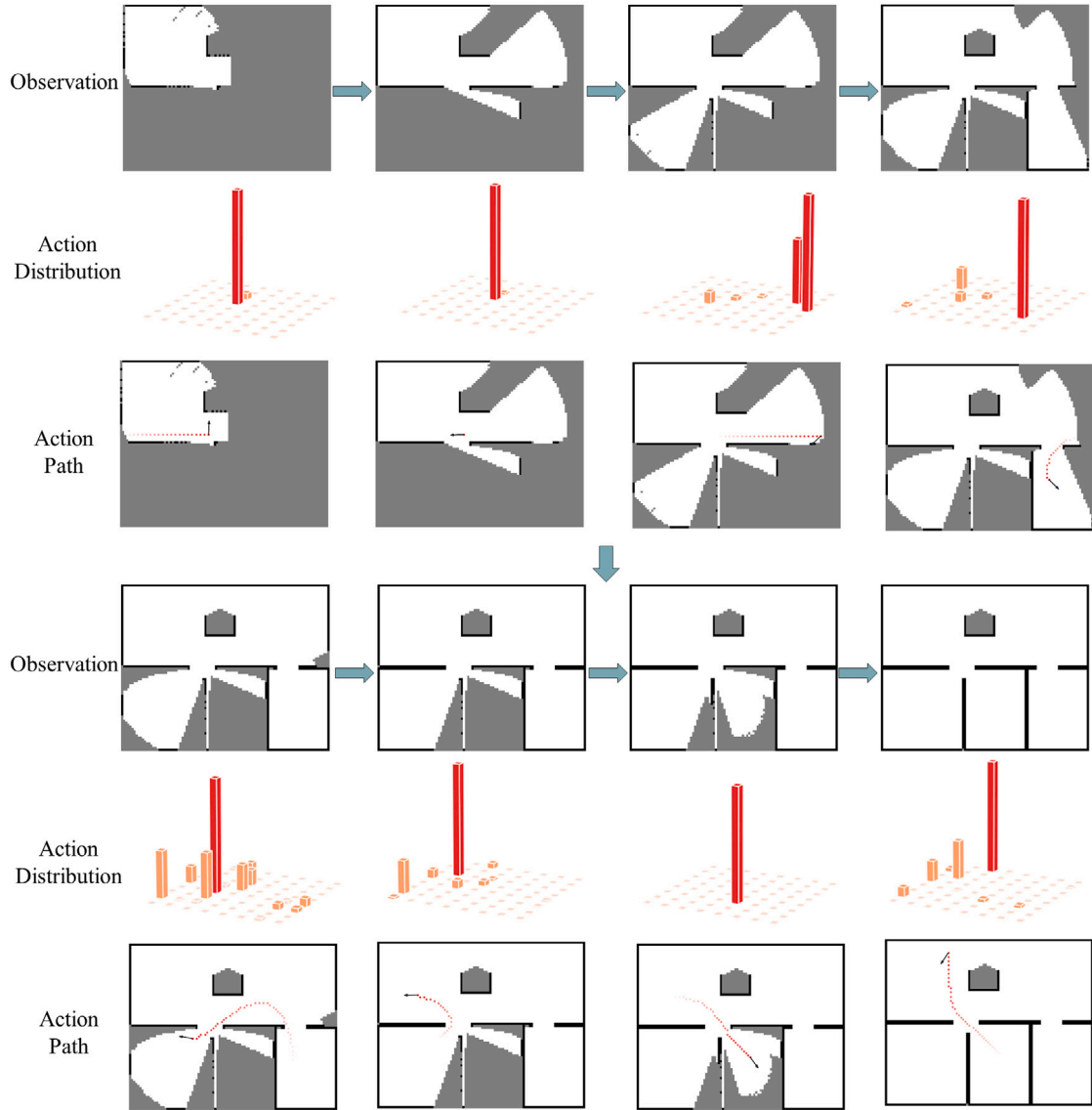


Fig. 10. Decision process of DLAE. Observation refers to the occupancy maps from the mapping module. Action Distribution represents the predicted probability of each position action. In action planning, red points represent the planning trajectory and the arrows are the sensing yaw.

within the Deep Learning Autonomous Exploration (DLAE) framework has yielded superior performance across all evaluated metrics, evidencing the significant strategic enhancements achieved. The results from the ablation experiments validate the efficacy of the proposed methods in augmenting the learning capability of UAV autonomous exploration models and in devising more effective exploration strategies.

In parallel, the ablation studies reveal that the lightweight model introduced herein, which possesses merely 1/10 the parameter volume of the original model, still attains commendable outcomes across all performance metrics. This lightweight model promotes quicker decision-making and alleviates computational demands on UAVs while maintaining acceptable performance standards, demonstrating its potential to further enhance the efficiency of UAV autonomous operations.

5.5. Decision process

This subsection visualized the decision-making process of DLAE. Our model predicts the distribution of the position action probabilities through observations and samples the distribution to determine decisions. Moreover, the planning and tracking modules implement the actions in practice. The whole decision process is illustrated in



Fig. 11. UAVs in Airsim Simulation Environments.

Fig. 10, which presents some exciting facts based on the decision process results. Specifically, our model selects to explore positions near unknown regions. However, unlike the frontier-based method, our model also focuses on environmental features such as walls and their position, which contribute toward a more efficient exploration.

5.6. Physical experiments

The proposed method is deployed on the Airsim simulation platform [37] to verify its performance in a more realistic environment.

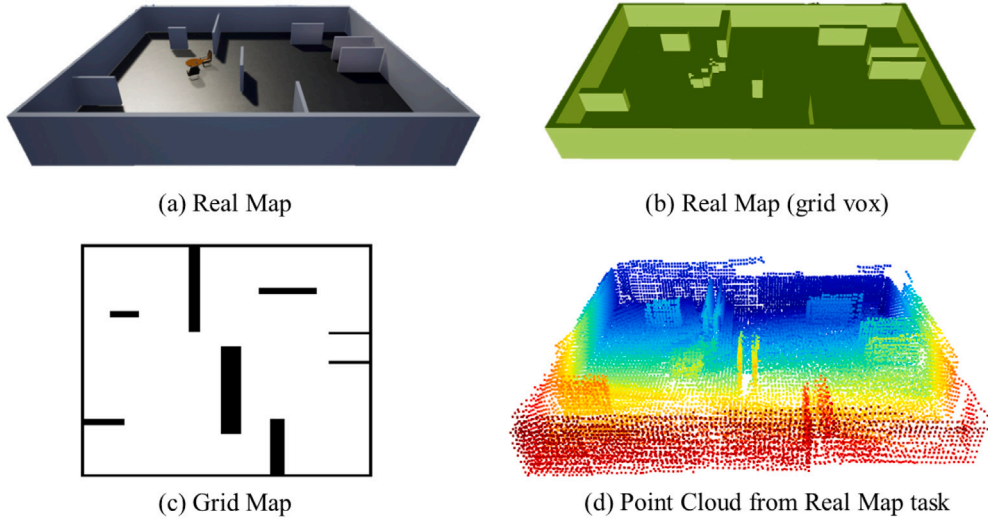


Fig. 12. Physical Experiments Results. The room in Airsim (a) and the real voxel grid of room (b). (c) is the grid map, and (d) is the reconstructed point cloud from the depth camera.

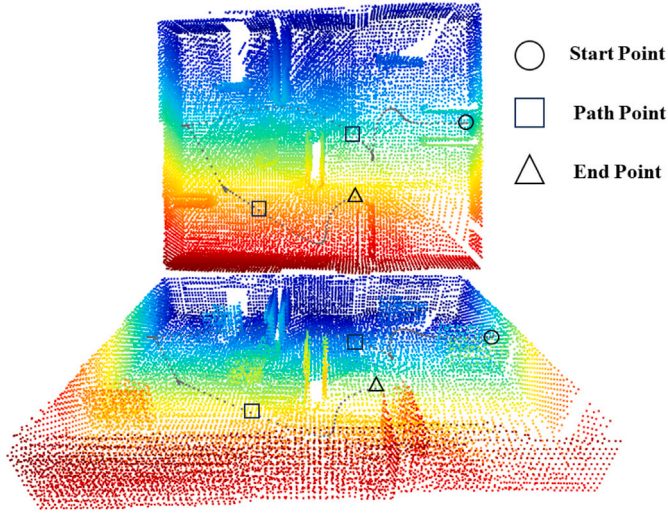


Fig. 13. A trajectory within the Airsim simulation. The circle indicates the starting point, the triangle represents the ending point, and squares represent path points.

Since the method's output is a minimum jerk trajectory, the trajectory is easy to track in practice.

Furthermore, we deploy the DLAE on the Airsim UAV depicted in Fig. 11. The UAV has a camera with 45-degree FOV and an effective depth range of 0.1 to 10 m, located at the front center of the UAV. The lift coefficient $C_T = 0.109919$, the maximum speed of the motor $\omega = 106.61112$, and the rotor length is 0.2286 m. Fig. 12 presents the UAV's exploration area and the final reconstructed point cloud. Moreover, the exploration trajectory in simulation is presented in Fig. 13. The DLAE completes the exploration of the room efficiently, proving its capability in a physical environment. The entire process will be showcased in the video.¹

6. Conclusion

Unmanned Aerial Vehicles (UAVs) demonstrate remarkable versatility in executing tasks across diverse and intricate environments, rendering autonomous exploration crucial for navigation within unfamiliar territories. Yet, autonomous exploration by UAVs encounters numerous hurdles, including delayed decision-making, Field of View (FOV) constraints, a proliferation of invalid actions, and an overly expansive action space. Addressing these challenges, this study introduces a deep learning approach, designated as DLAE, which integrates comprehensive modules for mapping, decision-making, planning, and tracking aimed at autonomous exploration. Distinctively, DLAE employs a neural network within its decision module, facilitating expedited and strategic global planning decisions, unlike traditional methodologies. The devised hybrid action space, comprising positional and yaw actions, innovatively mitigates FOV limitations, enhancing decision efficacy.

Our findings reveal that the primary impediments to autonomous UAV exploration are an unwieldy action space and a high incidence of invalid actions, which complicate the training process and obstruct effective policy development. To surmount these obstacles, we have designed an autoregressive network complemented by an invalid action masking strategy, significantly improving DLAE's performance metrics over conventional methods. Evaluations conducted across multiple test environments underscore DLAE's superior exploration rates and efficiency. Visualization of DLAE's decision-making process highlights its enhanced intelligence and efficiency. Additionally, empirical tests confirm DLAE's adeptness in real-world scenarios, illustrating its potential in facilitating complex tasks such as indoor mapping, area-specific target search, and semantic mapping construction.

In sum, DLAE notably enhances UAVs' capability to navigate and perform tasks in unknown environments more swiftly and efficiently. The application of DLAE enables UAVs to proficiently undertake complex assignments, including indoor map reconstruction, localized target identification, and the creation of semantic maps. Nonetheless, DLAE's exploration decisions are currently limited to a two-dimensional plane, omitting vertical mobility which could further enrich exploration strategies. Future research will, therefore, focus on incorporating three-dimensional decision-making capabilities and adaptable flight altitudes into DLAE, broadening the scope of autonomous UAV exploration.

¹ <https://youtu.be/j64l3CVpszl>

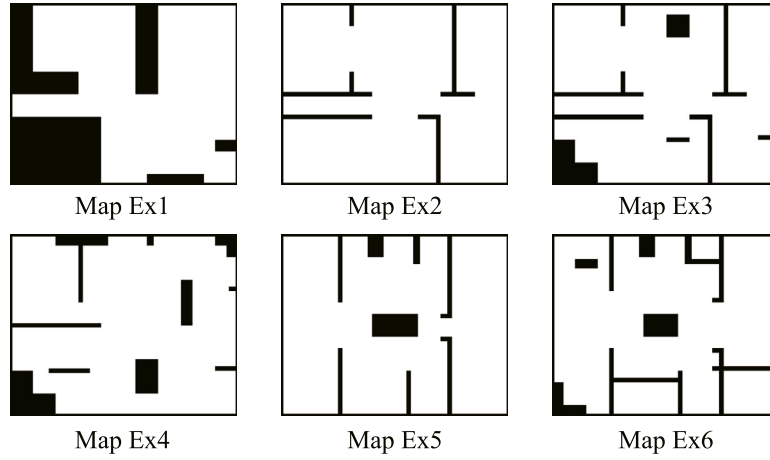


Fig. A.1. Maps of the Expanded Experiments. Map Ex1 is 4×5 meter. Map Ex2 to Ex4 is 8×10 meter. Map Ex5 and Ex6 is 13.4×16.6 meter. Some rooms in Map Ex6 are inaccessible.

	DLAE (ours)	DQN	PPO	Mask PPO	NBV	BO		DLAE (ours)	DQN	PPO	Mask PPO	NBV	BO
Ex1	0.737	0.521	0.738	0.728	0.743	0.740	Ex1	13.712	2.600	19.340	26.607	16.618	11.000
Ex2	0.965	-	0.923	0.947	0.995	0.996	Ex2	17.633	-	69.466	44.931	32.790	51.621
Ex3	0.903	-	0.871	0.907	0.961	0.948	Ex3	28.908	-	74.003	73.383	76.168	56.251
Ex4	0.902	-	0.831	0.843	0.932	0.931	Ex4	27.942	-	38.917	32.973	61.511	43.019
Ex5	0.849	-	-	0.545	0.977	0.970	Ex5	64.453	-	-	132.320	101.783	91.106
Ex6	0.751	-	-	0.332	0.806	0.817	Ex6	71.631	-	-	104.476	83.380	82.628

	DLAE (ours)	DQN	PPO	Mask PPO	NBV	BO		DLAE (ours)	DQN	PPO	Mask PPO	NBV	BO
Ex1	0.040	0.023	0.032	0.038	5.382	8.708	Ex1	0.054	0.200	0.038	0.027	0.034	0.038
Ex2	0.062	-	0.029	0.033	3.915	9.224	Ex2	0.055	-	0.013	0.021	0.027	0.016
Ex3	0.059	-	0.035	0.059	5.257	11.264	Ex3	0.031	-	0.012	0.012	0.012	0.014
Ex4	0.076	-	0.027	0.015	6.001	14.726	Ex4	0.032	-	0.021	0.026	0.014	0.016
Ex5	0.059	-	-	0.063	5.384	17.353	Ex5	0.013	-	-	0.004	0.009	0.009
Ex6	0.070	-	-	0.061	5.550	16.430	Ex6	0.010	-	-	0.003	0.009	0.008

(a) Explored rate

(b) Explored length

(c) Decision time

(d) Explored efficiency

Fig. A.2. Results of the Expand Experiments. A deep color means that the method performs well on the metric. The default is because DQN and PPO always select invalid actions on complex maps.

CRedit authorship contribution statement

Yumin Zhao: Writing – original draft, Visualization, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Jianlei Zhang:** Writing – review & editing, Validation, Data curation. **Chunyan Zhang:** Writing – review & editing, Supervision, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Expand experiments

To verify our method's effectiveness and generalization in more maps, we used six more different test maps, as illustrated in Fig. A.1. The corresponding performance is presented in Fig. A.2. Note that a deeper color indicates better performance on the metric. Moreover, some default data suggest that the method always selects invalid actions and cannot begin a valid exploration, such as DQN in Map Ex2 6 and PPO in Map Ex5 6. This is due to the poor generalization ability of these methods on complex maps. However, DLAE demonstrates an excellent generalization performance on maps of various complexity, achieving an outstanding performance on all metrics, especially in explored length and efficiency. This proves that DLAE is an efficient method for autonomous UAV exploration.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.knosys.2024.111925>.

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